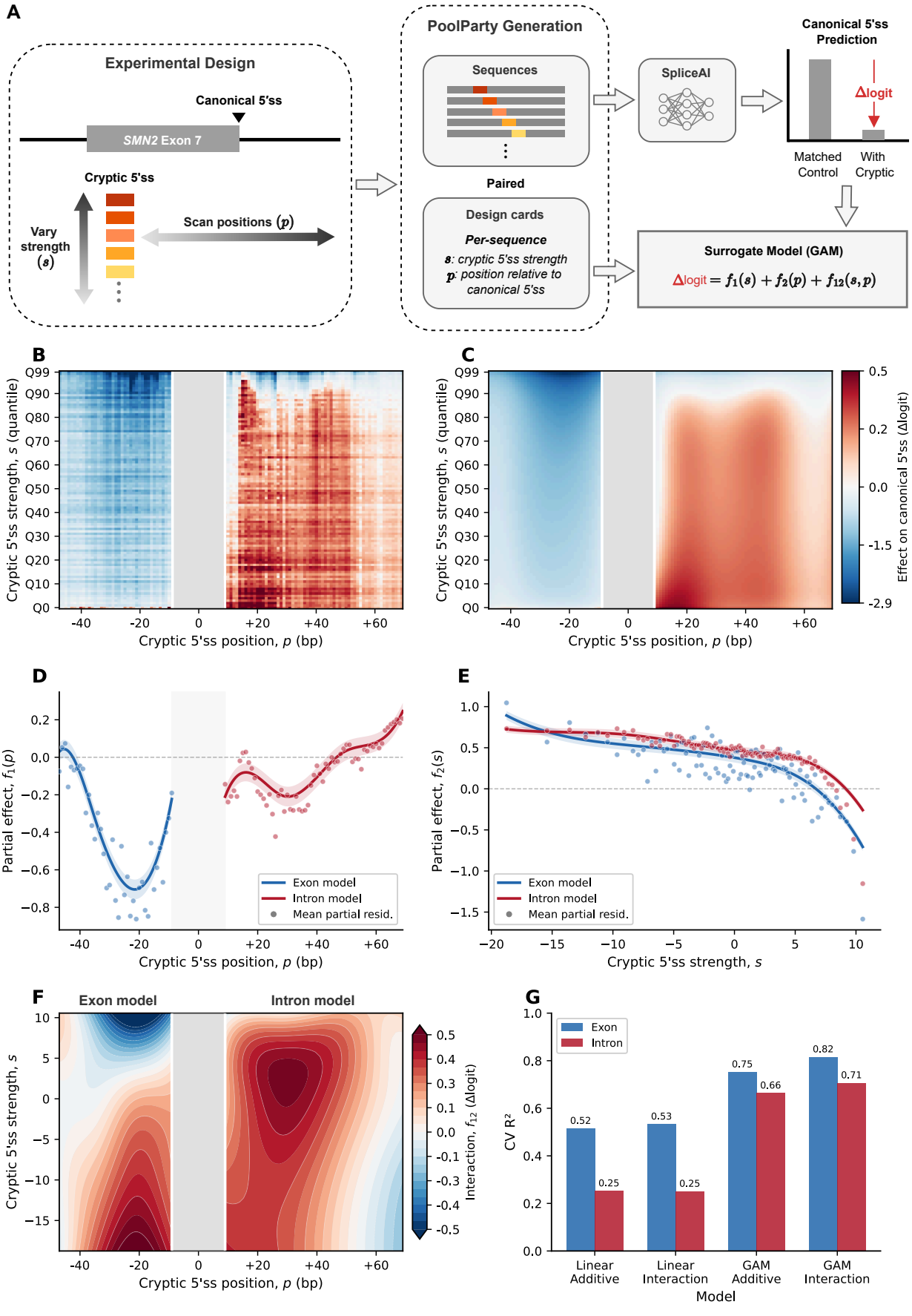
A
Experimental Design
Canonical 5'ss
SMN2 Exon 7
Cryptic 5'ss
Vary strength (s)
Scan positions (p)
PoolParty Generation
Sequences
Paired
Design cards
Per-sequence
s: cryptic 5'ss strength
p: position relative to canonical 5'ss
SpliceAI
Canonical 5'ss Prediction
Δlogit
Matched Control
With Cryptic
Surrogate Model (GAM)
Δlogit = f1(s) + f2(p) + f12(s,p)
B
Cryptic 5'ss strength, s (quantile)
Cryptic 5'ss position, p (bp)
C
Effect on canonical 5'ss (Δlogit)
D
Partial effect, f1(p)
Exon model
Intron model
Mean partial resid.
E
Partial effect, f2(s)
Cryptic 5'ss strength, s
F
Exon model
Intron model
Interaction, f12 (Δlogit)
G
CV R²
Exon
Intron
0.52
0.25
0.53
0.25
0.75
0.66
0.82
0.71
Linear Additive
Linear Interaction
GAM Additive
GAM Interaction
Model